Statistics (Set B) — MCQ Worksheet

Mathematics • Chapter 14 • *25 Questions, 1 mark each — Total: 25 marks*

Instructions: *Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.*

Q1. Mean of 4, 6, 8, 10, 12 is:

(A) 8 (B) 10
(C) 7 (D) 12

Q2. Median of 5, 9, 7, 6, 12, 8, 10 (when sorted) is:

(A) 7 (B) 8
(C) 9 (D) 10

Q3. Mode of 2, 3, 4, 4, 5, 6, 4, 7 is:

(A) 3 (B) 4
(C) 5 (D) 6

Q4. If mean of 5 numbers is 10, sum of numbers is:

(A) 50 (B) 5
(C) 10 (D) 15

Q5. Arithmetic mean is most affected by:

(A) Median (B) Mode
(C) Extreme values (D) Range

Q6. Class mark of class 15-25 is:

(A) 15 (B) 25
(C) 20 (D) 10

Q7. If Σf■x■ = 300 and Σf■ = 30, mean is:

(A) 10 (B) 100
(C) 3 (D) 30

Q8. For grouped data with classes of size 10, h in step-deviation method is:

(A) 10 (B) 100
(C) 1 (D) 5

Q9. In a frequency distribution, mode is:

(A) Value with smallest frequency (B) Value with highest frequency
(C) Middle value (D) Mean of frequencies

Q10. Mid-value of class 5-15 is:

(A) 10 (B) 5
(C) 15 (D) 7.5

Q11. Number of observations falling in class is called its:

(A) Class mark (B) Frequency
(C) Range (D) Mean

Q12. For an even number of data values, median is:

(A) Smallest (B) Largest
(C) Average of two middle values (D) Largest of two middle values

Q13. If mean of 6 observations is 8, mean of 5 of them is 7. The 6th observation is:

(A) 13 (B) 14
(C) 9 (D) 11

Q14. If median of x, x+2, x+4, x+6, x+8 is 5, then x is:

(A) 1 (B) 3
(C) 5 (D) 7

Q15. Empirical relation: Mode + 2 Mean =

(A) 3 Median
(B) 2 Median
(C) Mean
(D) Median

Q16. If mean of n observations x_1, ..., x■ is x■, mean of x_1+5, ..., x■+5 is:

(A) x■
(B) x■+5
(C) x■–5
(D) x■/5

Q17. Mean of first 10 odd natural numbers:

(A) 10
(B) 11
(C) 12
(D) 5

Q18. Median of 35 observations is the value of:

(A) 17th observation
(B) 18th observation
(C) 19th observation
(D) 35th observation

Q19. If mode is 65 and median is 55, mean (using empirical relation) is:

(A) 50
(B) 60
(C) 45
(D) 55

Q20. For grouped data, formula for median is l + ((n/2 – cf)/f) × h. Here, l is:

(A) Upper limit of median class
(B) Lower limit of median class
(C) Class mark
(D) Mode

Q21. If mean of 5 observations is 12 and mean of 7 other observations is 16, mean of all 12 observations is:

(A) 13.5
(B) 14.33
(C) 14
(D) 14.5

Q22. If x■ = a + h(Σf■u■/Σf■), where u■ = (x■ – a)/h. The formula assumes:

(A) Equal class size h
(B) Random class size
(C) No frequency
(D) a = 0

Q23. In a less-than ogive, the points plotted are:

(A) (upper limit, cumulative frequency)
(B) (class mark, frequency)
(C) (lower limit, cumulative frequency)
(D) (midpoint, mean)

Q24. If observations 4, 5, 6, 7, 8, 9 are increased by 3, the new mean is:

(A) 9.5
(B) 6.5
(C) 3
(D) 12

Q25. In a moderately skewed distribution, the value of mean is 32 and that of median is 30. The value of mode is:

(A) 26
(B) 28
(C) 30
(D) 32

— End of Worksheet —